Statistics 5810/6810: Graphics 1

Why are graphics discussed in this course?

- Good graphics today require the computer
- Visualization enters every step of the data analysis cycle
  - Data cleaning – are there anomalies?
  - Exploration
  - Model checking
  - Reporting results
- Plots can uncover structure in data that can't be detected with numerical summaries
- Important communication skill

Preview: Statistics 6560

- Advanced graphics discussed in Stat 6560 (Graphical Methods)
- Topics include:
  - Bad graphics (and how to avoid those)
  - History of graphics
  - Human perception & color
  - From 1-dim (categorical) data to high-dim data
  - Statistical maps
  - Interactive and dynamic graphics

R's graphics model

- There are two models in R – painter and object-oriented
- We will use the painter's model
- The other is easy to get started but hard to tweak
- Painter's model – start with a blank canvas, add/paint on it in multiple passes

Know your data types

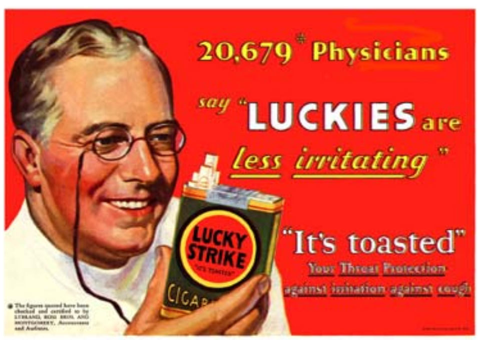
The appropriate graphical techniques depend on the kind of data that you are working with

- Quantitative
  - continuous – e.g. height, weight
  - discrete – numeric data with few values, e.g., number of children in family
- Qualitative
  - ordered – categories with an order but no meaningful distance between, e.g., number of stars for a movie rating
  - nominal - categories have no meaningful order, e.g., race

Case: Infant Health

load(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat5810/LectureNotes/KaiserBabies.rda"))

Smoking: Some doctors used to recommend it



Today:

SURGEON GENERAL'S WARNING: Smoking Causes Lung Cancer, Heart Disease, Emphysema, and May Complicate Pregnancy.

Kaiser Study

- Oakland Kaiser mothers
- 1960s
- Measure the babies weight (in ounces) at birth
- All babies:
  - Male
  - Single births (no twins, etc.)
  - Survived 28 days

Information collected on mother's and their babies

- Birth weight (ounces)
- Gestation (weeks)
- Parity - total number of previous pregnancies
- Mother's height and weight
- Mother's smoking status
- Mother's age, race, education level, income
- And more…

Distribution of Birth Weight

- The **distribution** is the pattern of variation in the birth weights.
- It provides the numerical values for birth weight and how often each value occurs.
- A **histogram/density plot** shows the shape of the distribution

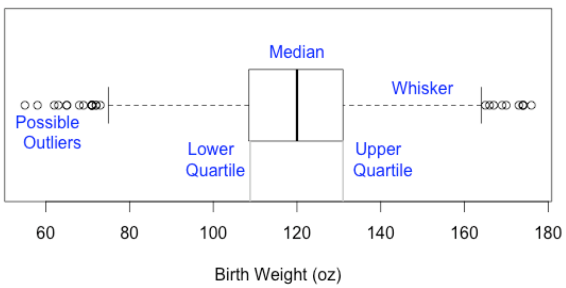
Histograms

- Are a special case of density plots
- AREA = Proportion (or percent)
- The area of a bar:

  Height * Width = Area

  (Proportion/oz) * oz = Proportion
- Histograms are not the same as bar charts
- With bar charts, it is only the height that matters. **Bar charts are for qualitative data**

Boxplot: `boxplot(infants$bwt)`

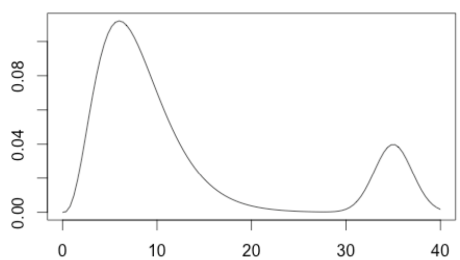


`boxplot(infants$bwt, …)`

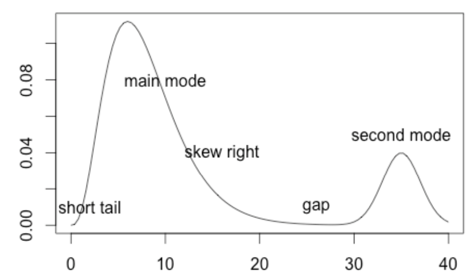
Looking for Structure: Quantitative Distribution

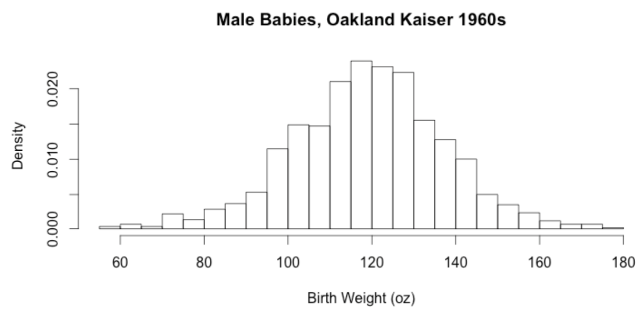
- **Distribution:** pattern of values for a variable
- **Mode**: high density region
- **Long Tail**: many observations far from center
- **Symmetry/Skewness:** distribution of values the left and right of the center.
- **Gaps**: places where there are no observations.
- **Outliers:** unusually large or small values that falls well beyond the overall pattern of data

What Structure Do You See?



What Structure Do You See?





Case: College Students

load(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat5810/LectureNotes/videogame.rda"))

STAT 2 Survey

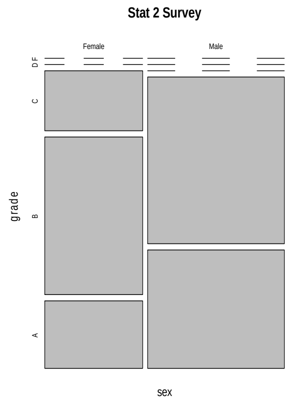
- Random Sample of 91 of 314 Cal students enrolled in Stat 2
- Survey collected the following info:
  - sex – Male/Female
  - grade – grade expected in the course (“A”, “B”, “C”, “D”, “F”)
- What type of data are these?
  - sex is qualitative (nominal)
  - grade is qualitative with an ordering

Method of Comparison

- Often, we not only want to better understand a distribution, but we want to compare the distribution for subgroups or to compare against another population or standard

- How do you think the expected grade distribution might vary with gender?

Two Qualitative variables



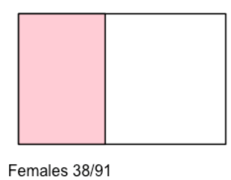
mosaicplot(table(video$sex, video$grade), main="Stat 2 Survey")

How to read a Mosaic plot

There are 91 students in the survey.
Think of them as spread out evenly in the box

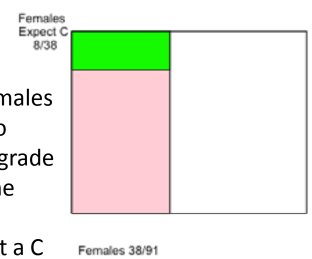
New Plot: Mosaic

Put all the females on one side of the box.
There are 38.

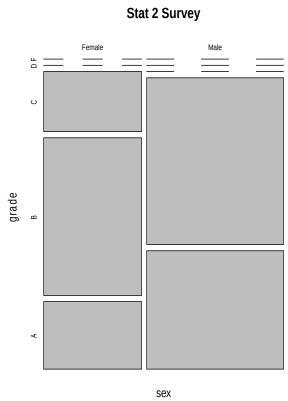


New Plot: Mosaic

Rearrange the females so that those who expect the same grade are together in the box.
8 of the 38 expect a C



Mosaic plot



None of the males expect a C

Smaller fraction of females expect an A in comparison to Males